Kadir Cihan Duran - PLSC 508 HW2

For this assignment I used a grooming network of bonobos from Torfs et al. (2023), “Multi-group analysis of grooming network position in a highly social primate,” published in PLoS ONE 18(4): e0284361. The authors observed bonobo groups in several European zoos. I used the largest group, which has 15 adult bonobos (10 females and 5 males, aged 7 to 53 years).

In this network, each tie shows how often two bonobos groomed each other, measured as the proportion of scans in which grooming was seen. The repository stores the network as undirected, so I used as it is. I also used three node attributes: sex, age in years, and rearing history. Thirteen bonobos were raised by their mothers, and two (ID40 and ID45) were raised in an unusual way, by humans, which is called as atypically reared. Almost every pair of bonobos grooms at least a little: 68 of the 105 possible pairs have some grooming (density 0.65). Because of this, I focused on regular grooming. I made a binary network in which two animals are tied if their grooming rate is above the median of all non-zero rates. This gives 34 ties. I show descriptive statistics of both networks in Table 1. I used this strong grooming network for CUG, ERGM and LSM, and as the main outcome in QAP.

Table 1. Descriptive statistics of the grooming network.

Network	Nodes	Ties	Density	Transitivity	Mean degree	Degree centralization
Any grooming (> 0)	15	68	0.648	0.670	9.07	0.242
Strong grooming (I analyzed)	15	34	0.324	0.370	4.53	0.286

Figure 1 shows the strong grooming network. It has 15 nodes and 34 ties, so its density is 0.32. The average animal has about 4.5 strong grooming partners. Transitivity is 0.37, and degree centralization is 0.29. The two atypically reared animals stand out in the figure. ID40 has only one strong tie, and that tie is to ID45, the other atypically reared animal. ID45 has three ties. Both animals sit at the edge of the network.

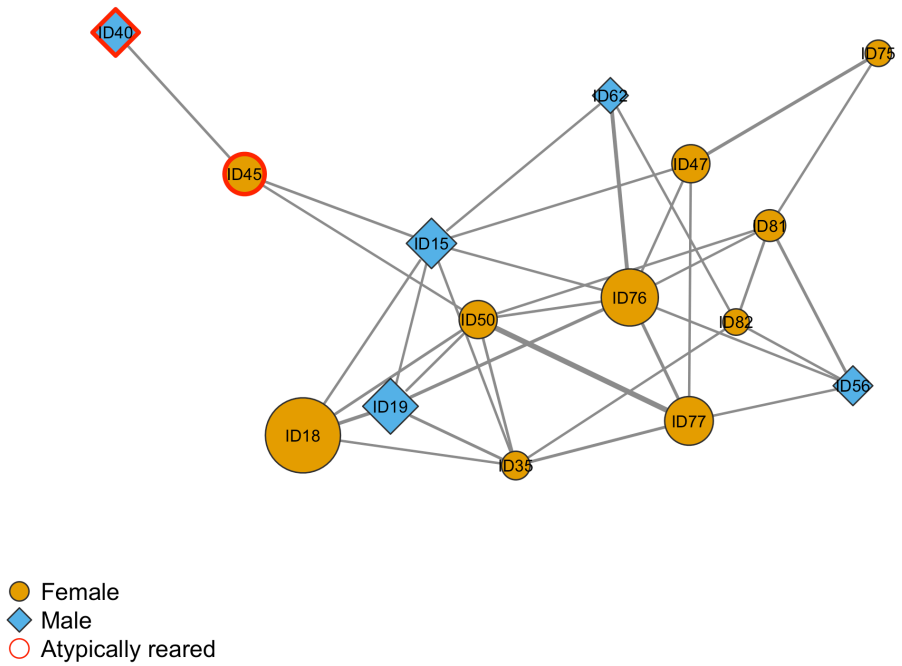


Figure 1. Strong grooming network. Circles are females and diamonds are males. Node size shows age, red outlines mark atypically reared animals, and edge width shows grooming rate.

I used conditional uniform graph (CUG) tests to see whether three network properties are different from what random networks would produce (Table 2 and Figure 2). For each test I made 1,000 random graphs.

Table 2. CUG test results (1,000 random graphs each).

Statistic	Conditioned on	Observed	Random mean	Pr(X ≥ Obs)	Pr(X ≤ Obs)
Density	Size	0.324	0.499	1.000	< 0.001
Transitivity	Number of ties	0.370	0.309	0.132	0.873
Degree centralization	Number of ties	0.286	0.252	0.498	0.848

Density: I compared the observed density (0.32) with random graphs of the same size, where each tie has a 50% chance. None of the 1,000 random graphs had a density as low as the

observed one ($p < 0.001$). This tells me that strong grooming ties are selective: each bonobo regularly grooms only some of the group. This result is expected, because I built the network using a median cut-off.

Transitivity: This is the more important test. Since denser networks also tend to have more closed triangles, I compared the observed network with random graphs that have the same number of ties (34). The observed transitivity is 0.37, and the random graphs have an average of 0.31. However, 13% of the random graphs were at least as transitive as the observed network ($p = 0.13$). So, I cannot say that grooming ties cluster more than chance.

Degree centralization: Again, holding the number of ties fixed, the observed centralization (0.29) is very close to the random average (0.25), with $p = 0.50$. This means no single bonobo has many more strong grooming partners than we would expect by chance.

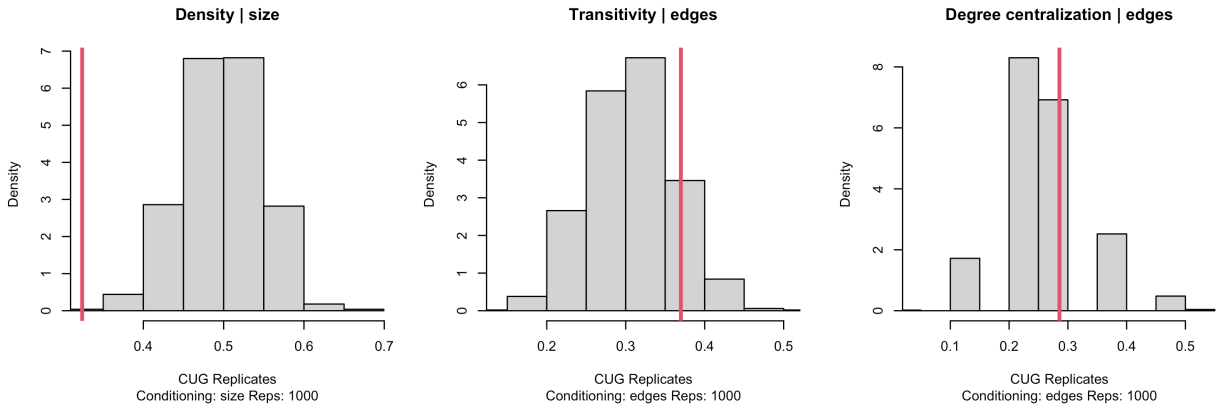


Figure 2. CUG null distributions for density, transitivity and degree centralization. The red line is the observed value.

I used QAP to test whether the grooming network is related to dyad-level covariates. I built five covariate matrices: (1) both animals are female, (2) both animals are male (mixed pairs are the reference group), (3) the absolute age difference between the two animals, (4) the sum of their ages, and (5) at least one of the two animals is atypically reared.

Table 3. QAP correlations between the strong grooming network and each covariate (1,000 permutations).

Covariate	Correlation (r)	Pr(≥ r)	Pr(≤ r)
Female-female	0.059	0.281	0.726
Male-male	−0.086	0.794	0.223
Age difference	−0.011	0.515	0.486
Age sum	0.114	0.165	0.835
Atypically reared (at least one)	−0.267	0.994	0.006

First, I tested each covariate on its own with QAP correlation. Only rearing was related to strong grooming. Pairs that include an atypically reared animal were less likely to be tied ($r = -0.27$, $p = 0.006$). The correlations for female-female pairs ($r = 0.06$), male-male pairs ($r = -0.09$), age difference ($r = -0.01$) and age sum ($r = 0.11$) were small and not significant. Figure 3 shows three of the permutation distributions.

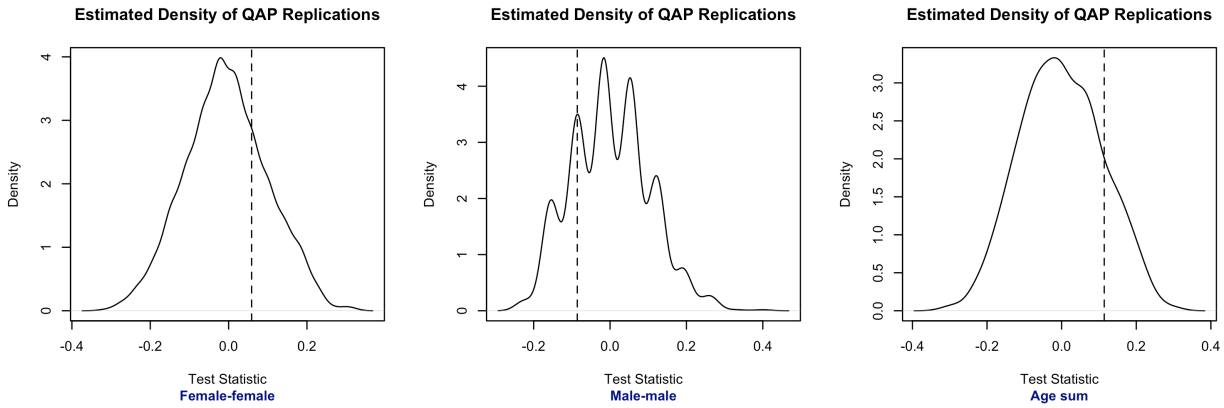


Figure 3. QAP permutation distributions of the correlation between the strong grooming network and female-female pairs, male-male pairs and age sum.

Next, I put all covariates into one QAP logistic regression (Table 4). Three effects were significant. Pairs with an atypically reared animal had about 85% lower odds of a strong tie (odds ratio 0.15, $p = 0.005$). Each extra year of combined age raised the odds of a strong tie by about 4% (odds ratio 1.04, $p = 0.040$), so older animals have more strong ties. Each extra year of age difference lowered the odds by about 6% (odds ratio 0.94, $p = 0.041$), so animals of similar age groom each other more. This age-difference effect did not show up in the simple correlation. It only appeared after I controlled for age sum, because the two age measures are related.

The sex effects were not significant. Female-female pairs were not more likely to be tied than mixed pairs (odds ratio 1.17, $p = 0.74$). Male-male pairs were less likely to be tied (odds ratio 0.59), but this was not significant ($p = 0.57$). There are only 10 male-male pairs, so this test has little power. The classical and QAP p-values were very similar, which suggests that the network dependence QAP protects against is not strong in this small group.

I also ran a QAP linear regression on the grooming rates themselves, without the cut-off. The age-sum effect ($p = 0.034$) and the rearing effect ($p = 0.012$) stayed significant. Pairs with an atypically reared animal groomed about one percentage point less often. The age-difference effect became weaker ($p = 0.10$). So, the main QAP results do not depend on how I made the binary network.

Table 4. QAP regression results. Logit: binary strong grooming network. Linear: grooming rate (% of scans), no cut-off.

Predictor	Logit coef.	Odds ratio	p (classical)	p (QAP)	Linear coef.	p (QAP)
Intercept	−1.235	0.29	0.055	0.060	0.334	0.460
Female-female	0.159	1.17	0.738	0.734	0.118	0.771
Male-male	−0.527	0.59	0.557	0.572	−0.567	0.377
Age difference	−0.058	0.94	0.047	0.041	−0.040	0.100
Age sum	0.041	1.04	0.038	0.040	0.036	0.034
Atypically reared (at least one)	−1.917	0.15	0.006	0.005	−0.985	0.012

QAP p-values are from 2,000 permutations.

Then, I fitted three ERGMs. Model 0 has only an edges term. Model 1 adds the same covariates as in the QAP models: nodematch("sex", diff = TRUE) for female-female and male-male pairs, absdiff("age") for age difference, nodecov("age") for age, and nodefactor("rearing") for atypical rearing. Model 2 adds two network terms to Model 1: gwesp(0.5, fixed = TRUE) for clustering and gwdegree(0.5, fixed = TRUE) for the spread of degrees. I used these geometrically weighted terms because simpler terms such as triangle might cause model degeneracy.

Table 5. ERGM estimates (standard errors in parentheses).

Term	Model 0	Model 1	Model 2
Edges	−0.74 (0.21)	−1.25 (0.63)*	−1.90 (1.08)+
Female-female (nodematch)		0.15 (0.47)	0.20 (0.49)
Male-male (nodematch)		−0.56 (0.88)	−0.60 (0.90)
Age difference (absdiff)		−0.054 (0.028)+	−0.057 (0.029)+
Age (nodecov)		0.039 (0.019)*	0.043 (0.021)*
Atypically reared (nodefactor)		−1.48 (0.61)*	−2.12 (1.00)*
GWESP (0.5)			0.11 (0.31)
GWDEGREE (0.5)			3.24 (3.35)
AIC	134.2	132.8	135.7
BIC	136.9	148.8	156.9

* $p < 0.01$, $p < 0.05$, + $p < 0.10$.

Model 1 has no network terms, so it is a logistic regression. Its estimates are almost the same as the QAP logit: age difference −0.054 (p = 0.055), age 0.039 (p = 0.042) and atypical rearing −1.48 (p = 0.015). The rearing coefficient is a bit different because nodefactor counts the number of atypically reared animals in a pair, while my QAP variable only marks whether there is at least one. The sex effects are again not significant.

When I added the network terms, the covariate effects stayed almost the same: age 0.043 (p = 0.043), age difference −0.057 (p = 0.052) and atypical rearing −2.12 (p = 0.035). This shows that these effects are not caused by unmodelled network structure. The GWESP coefficient was small and not significant (0.11, p = 0.71), which matches the CUG transitivity result: once I account for density and the covariates, there is no extra clustering. The GWDEGREE term was also not significant. AIC was lowest for Model 1, while BIC, which penalizes extra parameters more, preferred the edges-only model. I think this makes sense with only 105 dyads.

Model 2 converged with no warning. The MCMC trace plots (Figure 4) move up and down around zero with no trend, and the density plots are centered on the observed values. All Geweke tests had p-values above 0.44, and each statistic had an effective sample size of at least 2,155 out of 5,000 draws. Based on these checks, I am confident the estimation converged and the model is not degenerate.

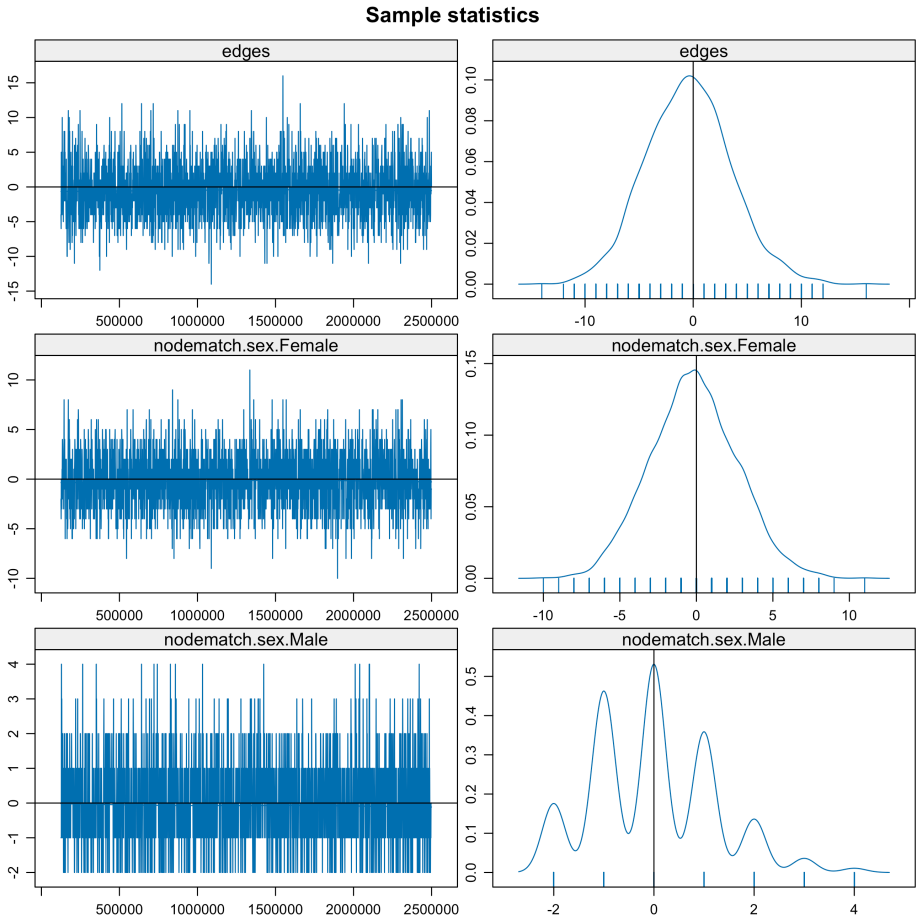


Figure 4. MCMC diagnostics for ERGM Model 2 (first three statistics; the other statistics look the same). Values are centered on the observed statistics.

I simulated 200 networks from each model and compared them with the observed network on the model terms, the degree distribution, edgewise shared partners and geodesic distances (Figure 5). For both Model 1 and Model 2, the observed values fall inside the simulated ranges, and no bin was significant at $p < 0.05$. Networks simulated from Model 2 also matched the observed transitivity (median 0.36 vs. observed 0.37). I conclude that both models fit the data well.

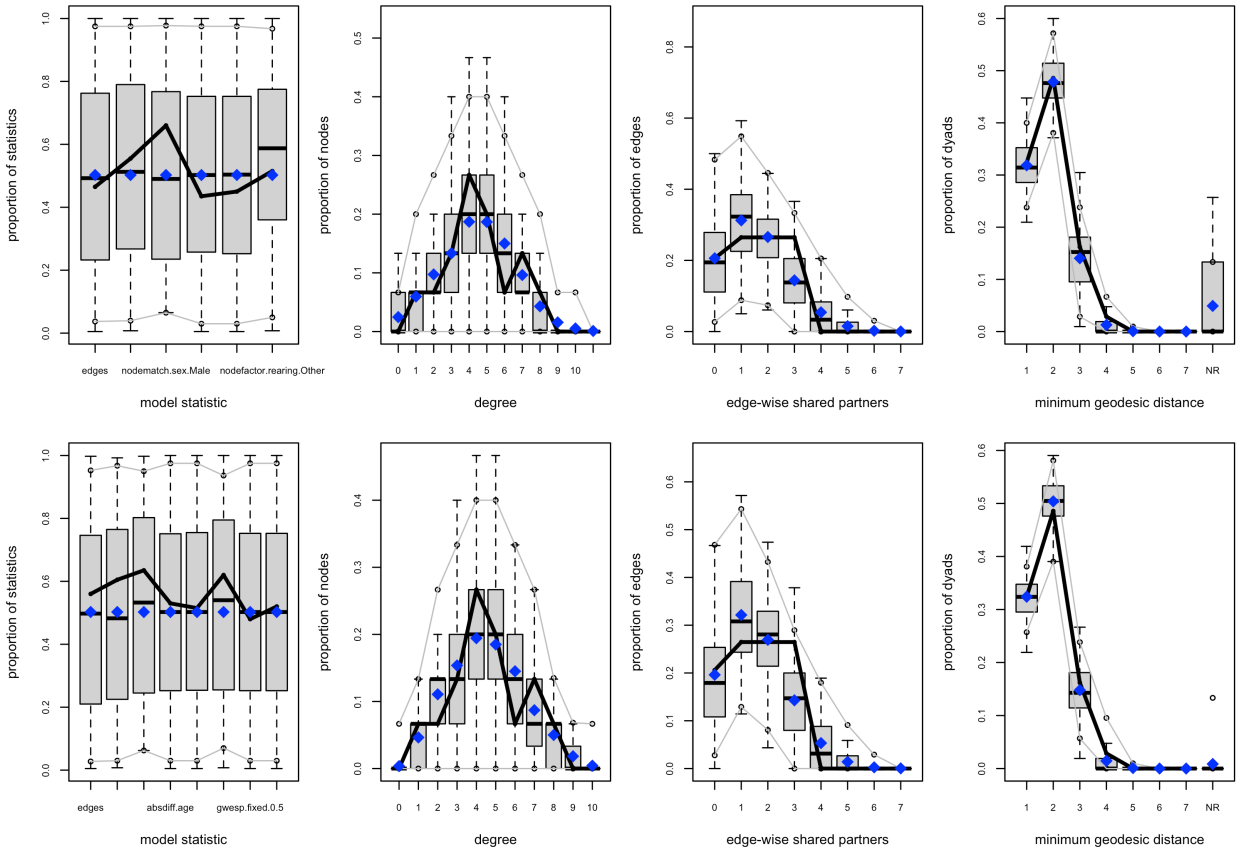


Figure 5. ERGM goodness of fit for Model 1 (top row) and Model 2 (bottom row). Black lines are the observed network; boxplots are 200 simulated networks.

Next, I fitted three Euclidean latent space models in two dimensions with latentnet. LSM 0 has latent positions only. LSM 1 adds the same covariates as before. LSM 2 adds a random sociality effect for each animal (rsociality). I used a burn-in of 100,000 iterations, a thinning interval of 50, and 4,000 saved draws.

Table 6. Latent space model estimates: posterior mean [95% credible interval].

Term	LSM 0	LSM 1	LSM 2
Intercept		0.79 [−1.03, 2.95]	1.00 [−1.39, 3.79]
Female-female		0.28 [−0.93, 1.48]	0.32 [−1.26, 1.93]
Male-male		−0.81 [−3.18, 1.21]	−0.89 [−3.67, 1.49]
Age difference		−0.054 [−0.125, 0.018]	−0.049 [−0.126, 0.030]
Age		0.042 [−0.006, 0.093]	0.039 [−0.022, 0.102]
Atypically reared		−1.70 [−3.43, −0.04]	−1.68 [−3.86, 0.60]
Sociality variance			0.68
Overall BIC	168.4	172.5	209.4

Table 7. LSM 1 convergence diagnostics.

Parameter	Effective sample size	Geweke z	R-hat (4 chains)
Intercept	2829	−0.12	1.000
Female-female	3302	−1.27	1.000
Male-male	3215	−0.89	1.000
Age difference	3283	−1.29	1.001
Age	2991	1.40	1.001
Atypically reared	3250	−0.14	1.000

The trace plots for the log-likelihood and the coefficients (Figure 6) show good mixing with no trend. The effective sample size was about 2,800 or more for every coefficient, and all Geweke statistics were between −1.3 and 1.4 (Table 7). I also ran four separate chains with different seeds, and the Gelman-Rubin R-hat was 1.00 for every coefficient. Together, these checks show the chains have converged.

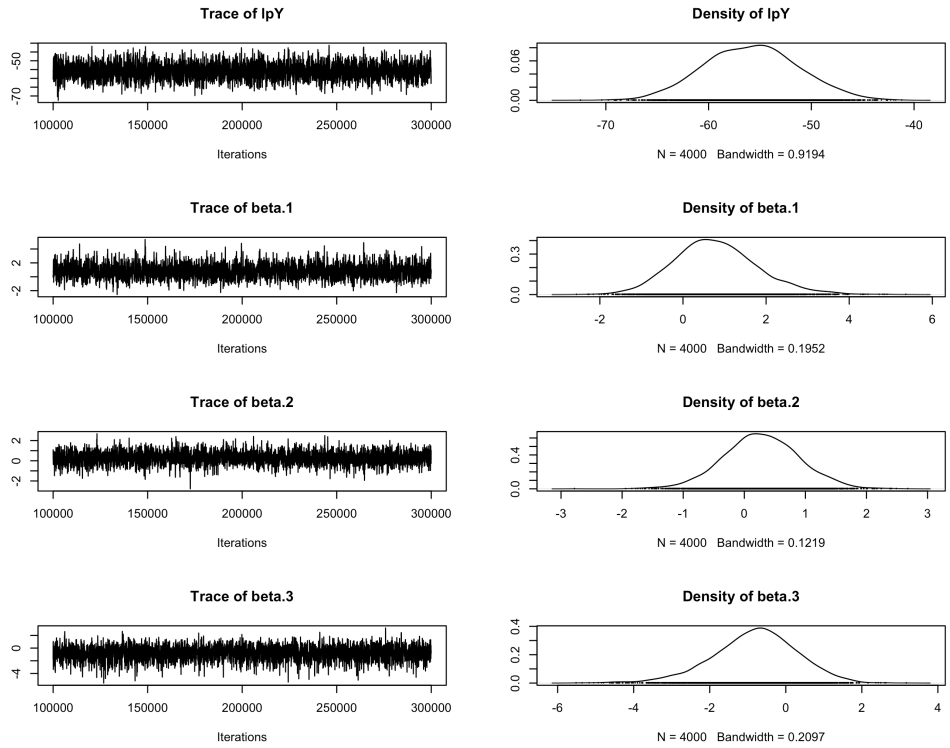


Figure 6. Trace and density plots for LSM 1 (log-likelihood and first three coefficients; the remaining plots look the same).

For goodness-of-fit, for 500 posterior draws, I simulated each pair once and compared the degree, edgewise shared partner and geodesic distance distributions with the observed network (Figure 7). No bin was significant at $p < 0.05$ for either LSM 1 or LSM 2. The simulated networks had the right number of ties (median 33). They had slightly higher transitivity than the real network (median 0.41 vs. 0.37), but the observed value was well within the 95% range. Distance models tend to do this, because if A is close to B and B is close to C, then A is also likely to be close to C. The in-sample AUC was 0.91 for LSM 1, compared with 0.73 for ERGM Model 1. However, this gain comes from 30 extra latent coordinates, so it does not mean the LSM would predict new data better.

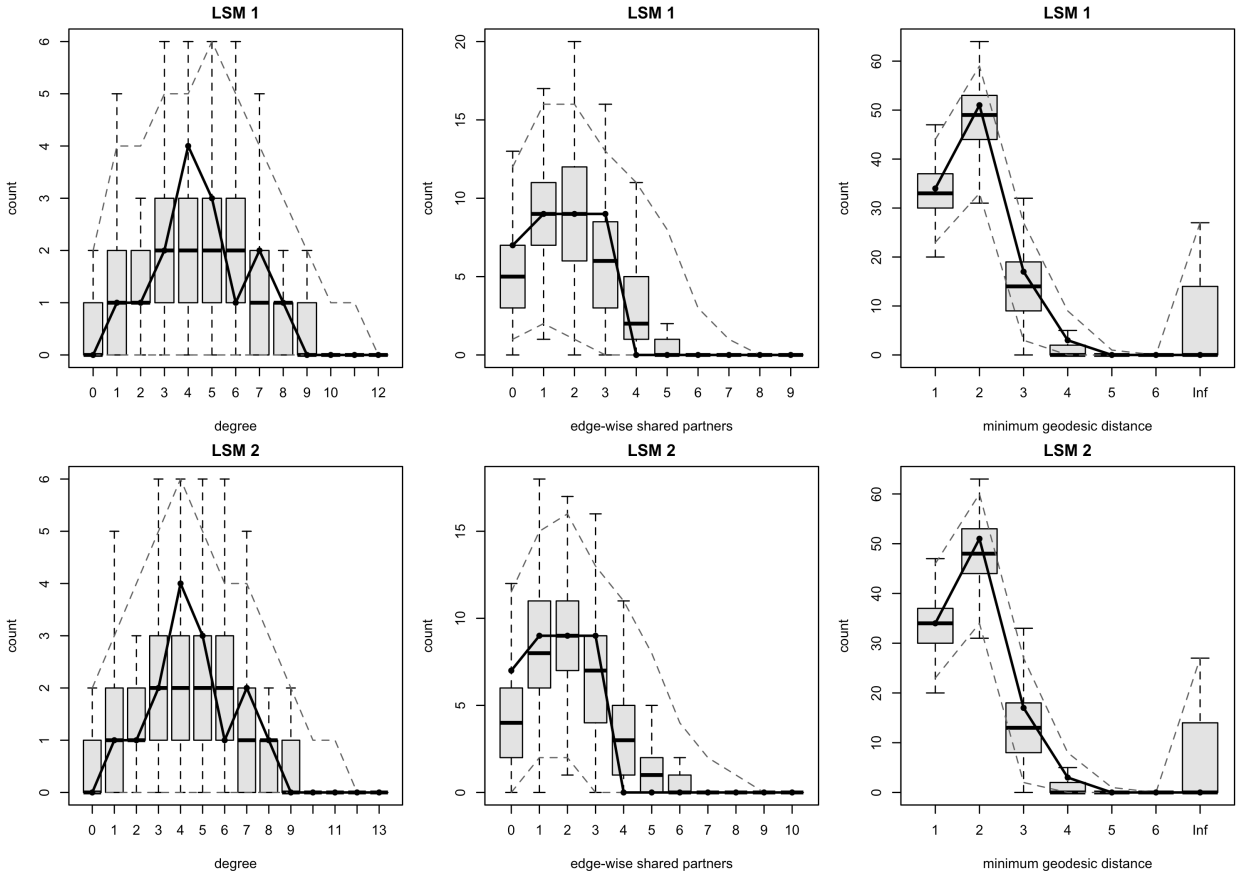


Figure 7. LSM posterior predictive goodness of fit for LSM 1 (top row) and LSM 2 (bottom row). The black line is the observed network; boxplots and dashed 95% bands come from 500 simulated networks.

BIC was lowest for the positions-only model (168.4), compared with 172.5 for LSM 1 and 209.4 for LSM 2 (Table 6). So, in a group of only 15 animals, the latent positions explain most of the pattern, and the covariates and sociality effects add little. In LSM 1, the covariate effects have the same signs and almost the same sizes as in QAP and ERGM: age 0.04, age difference −0.05, atypical rearing −1.70, female-female 0.28 and male-male −0.81. However, the 95% credible intervals are wider. Only the rearing interval does not include

zero, and only just (−3.43 to −0.04). The age intervals include zero. When I added sociality effects (LSM 2), the rearing interval also included zero (−3.86 to 0.60).

Figure 8 shows the estimated positions from LSM 1. The two atypically reared animals (ID40 and ID45) sit together at the edge of the space. They connect to the rest of the group only through ID45's ties to ID15 and ID50. The other animals form two loose areas. One area includes the oldest female (ID18) with ID19, ID15, ID35 and ID50. The other includes ID76, ID77, ID81, ID56, ID47 and ID75. Both areas contain females and males, which fits with the lack of a sex effect.

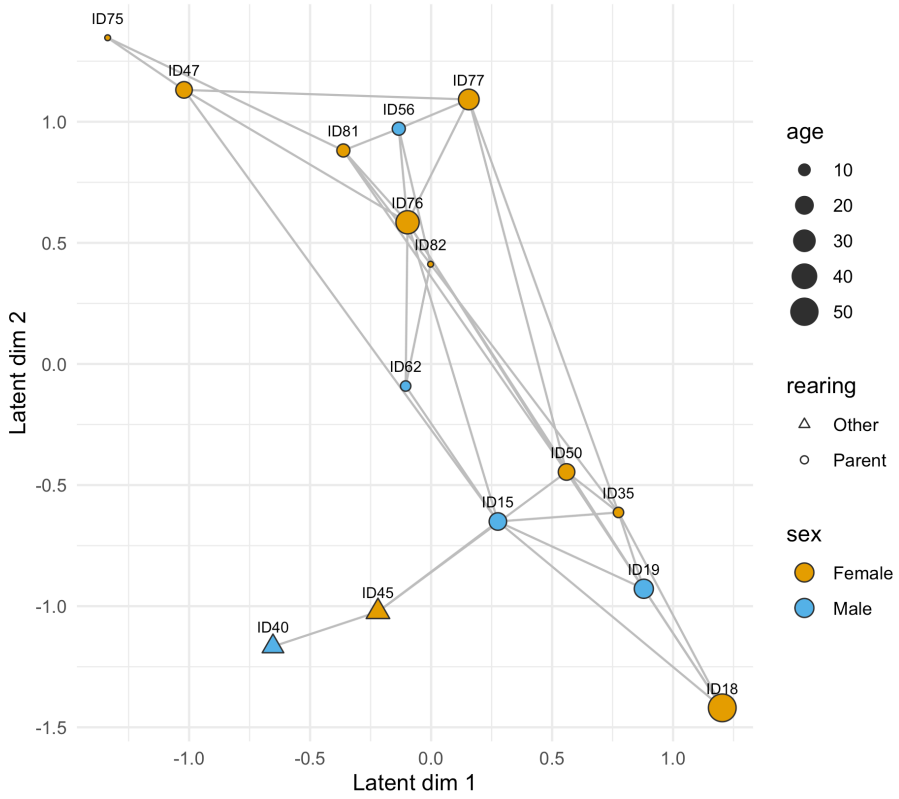


Figure 8. Latent positions from LSM 1. Grey lines are observed strong grooming ties. Animals that are closer together are more likely to be tied after accounting for the covariates.

Table 8 and Figure 9 put the results of all methods side by side. In this section I check where the methods agree and where they give different answers.

Table 8. Effects across methods. QAP and ERGM: coefficient (log-odds). LSM: posterior mean [95% credible interval].

Effect	QAP logit	ERGM 1	ERGM 2	LSM 1	LSM 2
Female-female	0.16	0.15	0.20	0.28 [−0.93, 1.48]	0.32 [−1.26, 1.93]
Male-male	−0.53	−0.56	−0.59	−0.81 [−3.18, 1.21]	−0.89 [−3.67, 1.49]
Age difference	−0.06*	−0.05+	−0.06+	−0.05 [−0.12, 0.02]	−0.05 [−0.13, 0.03]
Age (per year)	0.04*	0.04*	0.04*	0.04 [−0.01, 0.09]	0.04 [−0.02, 0.10]
Atypically reared	−1.92**	−1.48*	−2.12*	−1.70 [−3.43, −0.04]	−1.68 [−3.86, 0.60]
Clustering	–	–	0.11 (n.s.)	latent space	latent space

* p < 0.01, p < 0.05, + p < 0.10. LSM in bold: interval excludes zero. In QAP, age sum is equivalent to the ERGM nodecov(age) term.

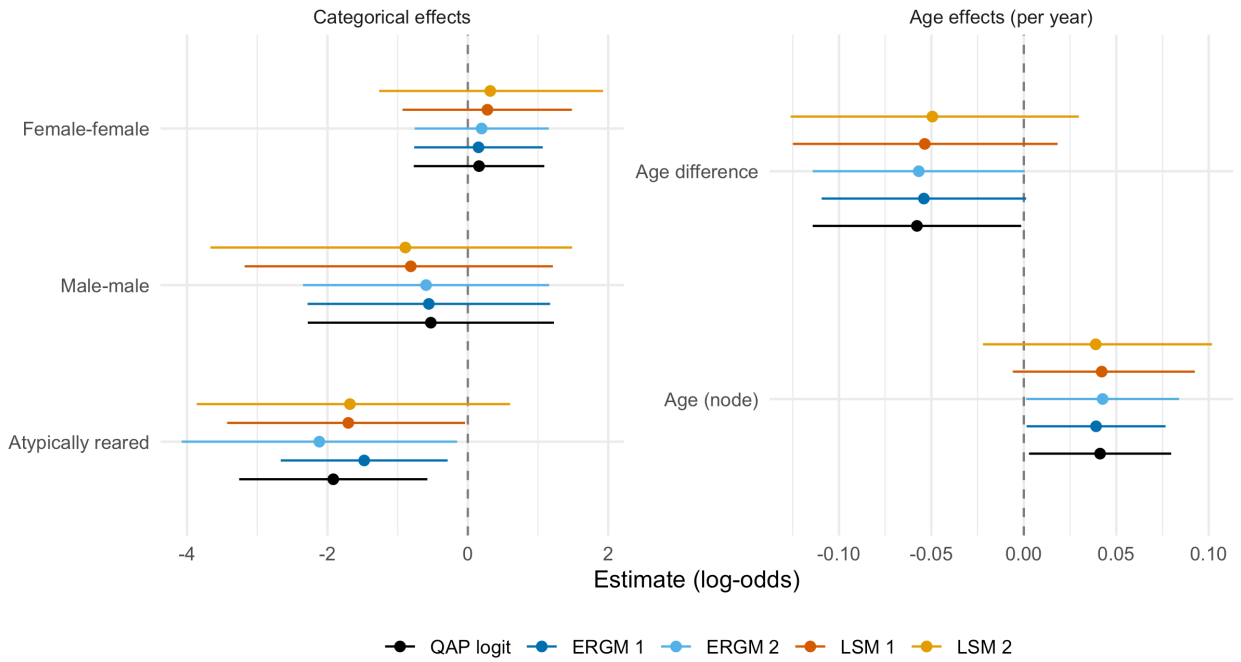


Figure 9. Covariate effects (log-odds) with 95% intervals across methods. QAP logit: classical standard errors; ERGM: Wald intervals; LSM: credible intervals.

Where the methods agree are the direction and size of the effects. QAP, both ERGMs and both LSMs give almost the same estimate for each covariate. Age is always positive, age difference is always negative, atypical rearing is always negative, and the sex effects are always small and not significant. The QAP logit and ERGM Model 1 are in fact the same

logistic model, so their estimates match. They differ only in how they compute p-values: QAP uses permutations, while ERGM uses model-based standard errors. In this network both give almost the same p-values.

Where the methods differ is statistical significance. QAP and ERGM Model 1 do not model network structure directly. ERGM Model 2 adds clustering and degree terms, and the LSM adds a latent position for every animal. As I move from QAP to ERGM to LSM, the uncertainty grows. The rearing effect is significant in QAP ($p = 0.005$), in both ERGMs ($p = 0.015$ and 0.035) and in LSM 1, but not in LSM 2. The age effects are significant in QAP and ERGM but not in the LSMs. I think this happens because, with only 15 animals, the latent space can explain some of the same pattern as the covariates. For example, it places the two atypically reared animals together and away from the core, so the rearing covariate has less left to explain.

Network structure is that the methods agree. The CUG test found no extra clustering ($p = 0.13$), and the GWESP term in ERGM was not significant ($p = 0.71$). Both ERGM Model 1 and the LSM reproduced the observed level of clustering without any special clustering term. In the same way, the CUG centralization test, the GWDEGREE term and the small LSM sociality effect all found no special popular animals. So, in this group, regular grooming seems to be shaped mostly by the traits of the two animals (age and rearing) rather than by friend-of-a-friend processes.

In summary, the results are consistent across methods in the following ways: Sex: no method finds a sex effect. Female-female pairs are never more likely to be tied than mixed pairs. Age: QAP and both ERGMs find a significant effect. The LSMs give the same direction, but the effect is not significant. Rearing: this is the most consistent result. It is significant in QAP, both ERGMs and LSM 1, and it has the same direction in LSM 2. Clustering and centralization: CUG, ERGM and LSM all agree that there is no extra clustering and no especially popular animal at the main cut-off.

My results also fit well with the original paper. Torfs et al. found that atypically reared bonobos had lower grooming centrality and that age mattered for grooming. I reached the same conclusions with methods that model the ties between animals directly, rather than summary scores for each animal.

Bonus Part

Because my binary network depends on a cut-off, I repeated the CUG transitivity test, the QAP logit and ERGM Model 1 with four cut-offs (Table 9).

Table 9. Results under different tie cut-offs.

Cut-off	Ties	Transitivity	CUG p	QAP FF	QAP MM	QAP age	ERGM MM	ERGM age
Any grooming (> 0)	68	0.670	0.082	−0.06	−1.15	0.03	−1.17	0.03
Above 25th percentile	51	0.514	0.144	0.10	−0.86	0.02	−0.88	0.02
Above median (main)	34	0.370	0.132	0.16	−0.53	0.04*	−0.56	0.04*
Above 75th percentile	17	0.441	< 0.001	−0.09	−16.87	0.05*	−Inf	0.05*

FF = female-female, MM = male-male. * $p < 0.05$. CUG p is $Pr(X \geq Obs)$ for transitivity, conditioned on the number of ties.

For age, the effect was positive at every cut-off and became significant from the median cut-off upward. So older animals mainly have more of the stronger grooming ties.

For male-male, the effect was negative at every cut-off. At the 75th-percentile cut-off there were no male-male ties at all, so the estimate goes to minus infinity (the ERGM reports −Inf and QAP reports about −17). This shows that males almost never have very strong grooming ties with each other. It also means I cannot estimate this effect at the highest cut-off.

For female-female, the effect was close to zero at every cut-off, so the lack of a female-female effect is robust.

For transitivity, this result did depend on the cut-off. Transitivity was not significant for the first three cut-offs (p between 0.08 and 0.14). But the 17 strongest ties were clearly clustered (transitivity 0.44, $p < 0.001$). So, the very strongest grooming bonds may form small, closed groups, even though regular grooming in general does not.